



Driving Knowledge Acquisition via Metric Life-Cycle in Process Intelligence

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ABSTRACT

Recently, much research work has focused on the design of complex infrastructures for Business Process Management (BPM).

Process Intelligence (PI) has emerged as a branch of the BPM domain that deals with improving and standardizing process inspection and analysis, aiming at improving the speed and the effectiveness of business operations. Currently, PI shows several limitations, e.g. due to the fact that only a limited part of the information collected by BPM systems is actually used.

The novel idea of the metric life cycle implemented in our approach is based on the definition of “check points”, allowing to better understand the role that each metric can or should cover within a BPM process, and providing an important upgrade in the assessment definition.

Categories and Subject Descriptors

H.4 [Information Systems Applications]: Miscellaneous;
D.2.8 [Software Engineering]: Metrics—*complexity measures, performance measures*

General Terms

Design, Performance

Keywords

Business Process Monitoring, Metric Life-Cycle, Metric Meta-model

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1. INTRODUCTION AND PROBLEM DESCRIPTION

Business Process Management (BPM) [12] represents today a wide discipline that combines knowledge from information technology and from management sciences; it embraces all techniques supporting design, enactment, execution, monitor and audit of business processes and process-aware information systems. Within BPM, Process Intelligence (PI) has emerged as a branch of process analysis focused on improving and standardizing the inspection of processes to provide deeper insights than traditional performance monitoring and measurement¹. PI applications are typically focused on providing real-time access to data about business activity or business process, with specific attention to critical business performance indicators, as also reported in the literature [4, 6, 8]. The aim is improving the speed and effectiveness of business operations. From a technological point of view, PI requires the convergence of operational business intelligence [9] and real-time application integration enabled through the advance of the Information Technology (IT). A known limitation of BPM applications today is that, as demonstrated by several studies, only a limited part of the information auxiliary to the business process execution is actually used [11].

According to the literature the most important capabilities that are covered by BPM are the following.

- **Analysis and discovery:** supporting the analysis of completed process instances from both a business and an IT perspective and also to discover process models. It helps analysts to identify the causes for a particular process behavior.
- **Monitoring:** analyzing process under execution. This can either be in the sense of alerting the user that some

¹Actually a large variety of buzzwords like Business Activity Monitoring (BAM), Business Operations Management (BOM), Business Process Intelligence (BPI), Process Mining, and Business Operations Intelligence (BOI) exists in the area, and it is not trivial to define distinctions among all of them. In this work we decided to focus on Process Intelligence (PI) as this term is connected with real-time access to data and analysis of auxiliary data.

unusual or unwanted situation is (about) to occur, or in the sense of interacting with the business process management system to prevent the violation of compliance, security, and contractual policies.

- **Optimization:** based upon information about past executions, identifying areas of improvement in business process definitions and in the resource allocation, so that, for instance, bottlenecks or exceptional behaviors do not happen.
- **Prediction:** based on the discovery capabilities, deriving prediction models and apply such models to running processes, to prevent the possibility of exceptions, undesired behaviors, and non-compliance.

However, the effort spent by the scientific community in developing these capabilities is not equally allocated [10, 11]. Actually, a major part of the current efforts are devoted to the design of complete infrastructures for BPM. For instance, as reported in [10], several studies focused on designing and enacting models, discover models from event data, selecting models from collections, composing models and so on, while optimization and improvement were less considered. This lack of attention is rooted in the origins of BPM that distinguished itself from Database Management by taking a control-flow perspective on Information Systems [7]; but persists in the today literature.

PI emerged in the last years as a research program aimed at maximizing the value of the information available in processes. In this paper, we claim that the full integration of BPM capabilities in a process model requires to be based on the notion of Metrics. In BPM metrics are measures of quantitative assessment used for descriptive statistics, comparison or to track process performances. Metrics are a critical element to capture the information carried out in a business process [3]. However, metrics are contextual in nature, and for this reason PI frameworks often lack in providing procedures for driving their definition and life-cycle. It is not surprising that a recent research involving 700 process professionals, conducted by Forrester Research in conjunction with PEX Network, found inadequacies with current BPM metrics employed by many organization [2]. Most of the current studies on PI focus on the analysis of the process behavior and result on performance improvement limited to this aspects [5].

This work is rooted within the activities of the KITE.it project for improving processes in the aerospace sector [1]. As described in Section 2 KITE.it gives particular attention to metrics and then life-cycle. In fact, the KITE.it framework should be capable of supporting procedures such as *i)* creation, contextualization and execution of metrics, *ii)* connection between metrics and analysis paradigms, and *iii)* visualization of the results. The final goal is driving the monitoring process by following a logic of knowledge discovery, to derive previously unknown and potentially useful knowledge.

For this purpose, we introduced a “metric life-cycle” model supporting the evolution of metrics from descriptive, to prescriptive, to inductive intention. More precisely, in our approach, metrics support the evaluation of process instances

based on the comparison with policies and KPI, defined at the strategic level, and exploit the inferences that can be generated from these comparisons in order to evolve metrics along the different steps of their life cycle.

The paper is organized as follow. Section 2 proposes an introduction to the KITE.it methodology. The metric definition workflow and the novel concept of “metric life-cycle” introduced in this paper are then presented and discussed in Section 3, allowing to give important considerations about the model proposed. Concluding remarks are finally reported in Section 4.

2. THE KITE METHODOLOGY

KITE.it is a project co-funded by the Italian Ministry for Economic Development, within the “Industria 2015” Programme, in the area of “New technologies for Made in Italy”. In this context, KITE.it is aimed at developing a methodological and technological framework to support the evolution of the aerospace supply chains towards value network models. In other words, the aim of the project is:

- supporting interoperability and cooperation among business networks and knowledge workers,
- monitoring the intellectual capital created and exchanged in the network, and
- measuring and optimizing performances.

In order to achieve these objectives, the project provides an environment for business and social cooperation that enables interoperability and cooperation among enterprises and other knowledge workers, making available all methodological and technological tools developed in the project. The KITE.it environment includes:

- *A general methodology (Kite Methodology)* to support the activities of analysis, design and monitoring, needed for the transition towards value network models. This methodology, shown in Figure 1, manages iteratively the entire business life cycle both at strategic and operational level. At the strategic level the exogenous variables and the value network in which the organization operate are analyzed. At the operational level, the strategy and the corporate policies are realized by an architecture of core processes.
- *A design environment*, reported in Figure 2, that includes tools and languages for process and IT services design. The system includes an integrated metamodel for the multidimensional business process modeling and a modeler to design the different business models or diagrams, that have been identified in order to describe the process architecture, the organizational structure, processes at various levels of detail, the business policies and the operational risks associated with the processes. In particular the integrated metamodel includes a model for managing metrics.
- *A Business Performance Evaluation Environment*. The system includes the definition of strategic and operative metrics, metrics on policies and risks (violations,

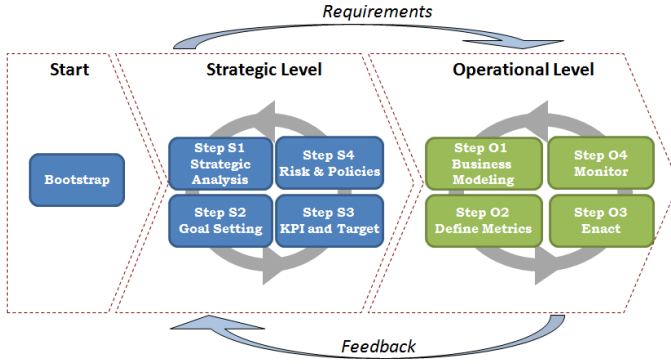


Figure 1: A diagram illustrating the Kite Methodology .

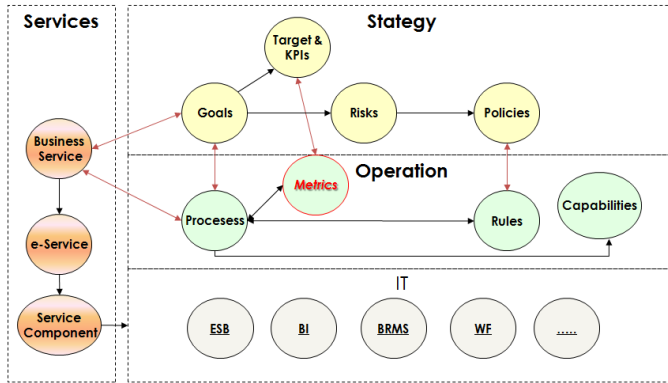


Figure 2: The design environment including tools and languages for process and IT services design.

dysfunctional behavior), security and Social Network Analysis (SNA) metrics and the development of a dashboard for their visualization, in order to monitor and optimize performances of the value network, improving the knowledge on the business process or on the business model, by means of the information about process, security and SNA measures.

The goal is driving the monitoring process by following a logic of knowledge discovery composed of four steps: (i) model knowledge; (ii) acquire data (iii) generate inference; and (iv) derive previously unknown and potentially useful knowledge. To do this the KITE.it framework provides a set of tools and services in order to map all monitoring assets. The user is able to map easily business models and metadata to existing and heterogeneous information artifacts generated or collected at the various phases of the methodology by means of Business and Social Cooperation Environment.

3. THE KITE METRIC SYSTEMS

3.1 Metric Definition Workflow

The metric process definition is strictly correlated both to the strategic and operational levels, as shown by the main components of a metamodel in Figure 1. In fact all the KITE.it metrics are monitor and check oriented: a general

process is defined at strategic level through the goals definition, but it is then validated only over the computational data. The most important step of such a process definition is represented by the metric definition, carried out in the KITE.it integrated system by the definition of the corresponding metric descriptors.

Nevertheless, the crucial steps of such a process regards the definition of target and the KPI, with risks and policies defined at strategic level. Indeed, the aim of a metric corresponds to the achievement measurement of such KPIs, targets, risks or policies. Moreover, during the metric definition some procedures are needed, as those related to the computation and data access, that will allow the consequent metric execution.

The metric definition step generally refers to a group of metamodel specifications aiming the final metric implementation. Such a phase involves metric descriptors modeled according to an XML Schema defined in the KITE.it metric metamodel. In particular, in order to satisfy all the requirements needed for the metric insertion, such a metamodel is divided into the following four main areas:

Measurement: it describes the network with strategic objectives (target, policies). Such a paradigm also allows to refer to the analysis paradigm for the objectives description.

Observation: it describes the metric observation conditions in terms of process instances or validity time of the data that will be analyzed. It also allows to define thresholds or validity time for critical metric values, like those indicating the success in an objective achievement.

Computation: it represents the operations that have to be carried out over the elements analyzed by the metric. It also describes the data access procedures, in terms of localization of the data sources required for the metric computation.

Visualization: it represents the structures that will be used for data visualization.

3.2 Metric Life-Cycle

In order to understand the role that metrics can play within the BPM process, we need to classify them according to their life-cycle. In particular, the metrics considered in our approach can be classified into three main categories:

- *Descriptive:* describe the process data and the behavioral attributes; give a first analysis based on them.
- *Prescriptive:* evaluate the objective achievements, as well as KPI and policies violations.
- *Inductive:* aimed at acquiring novel knowledge on the process through the evaluation of the statistic significance of the assumptions.

The first two categories are generally used to perform correlation analysis giving rise to the third metric category: the inductive metrics. This approach also corresponds to our “metric life-cycle”, which defines the different roles metrics

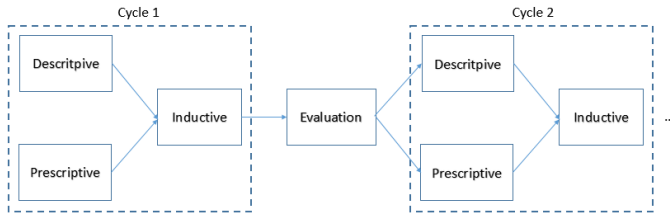


Figure 3: Diagram of our proposed metric life-cycle.

can play and defines development steps, allowing to classify the metrics belonging to the first two categories according to the results obtained after correlation analysis. Figure 3 shows an example of such a metric life-cycle diagram.

The main idea of the metric life-cycle is based on the definition of “check points”, allowing to better understand the role that each considered metric can cover within a BPM process, and providing an important feedback to the monitoring process.

This definition allows to clarify how our metric life-cycle check the evolutions steps from one cycle to the other. A metric that is defined to monitor descriptive policies and KPI can be considered prescriptive if it allow to detect violations. Then descriptive and prescriptive metrics can be used to derive inductive metrics that can be validated by statistical analysis. When statistical analysis validate an inference inductive metrics can start assuming a prescriptive intention, giving rise to a new cycle.

4. CONCLUSIONS

The literature has shown how the effort spent by the scientific community in introducing new modeling languages is out of all proportion with the small effort spent in developing systems for processes improvement, and most of the current studies are oriented on the design of the whole infrastructure able to cope with BPM.

In this paper we put forward the idea that the full integration of BPM capabilities requires to keep the notion of Metrics central, and that the value of the information available in processes becomes one of the most important topics in this area, reaching maximum values. In BPM Metrics are measures of quantitative assessment used for descriptive statistics, comparison or to track process performances. Metrics are then a critical element to capture the information carried out in a business process. However metrics are contextual and for this reason BPM frameworks often lack in providing procedures for driving their definition and life cycle.

The main idea of the metric life cycle is based on the definition of the so-called ‘check points’, allowing to better understand the role that each considered metric can or should cover within a BPM process, and providing an important upgrade in the assessment definition.

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